

Paper Replication Report #094: Magnetospheric Accretion Truncation & Inner Disk Cavity Radii

Antigravity Solar System Dynamics Replication Campaign

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Abstract

We present a first-principles replication of Ghosh & Lamb (1979), Koenigl (1991), Shu et al. (1994), and Bouvier et al. (2007) modeling magnetospheric interaction between Classical T Tauri Stars (CTTS) and gas disks, magnetic pressure truncation at the Alfvén radius $r_A \propto B_*^{4/7} \dot{M}^{-2/7}$, corotation radius r_{co} , and magnetospheric funnel flow accretion streams onto stellar magnetic poles. Using our C++ solver engine, we evaluate inner disk truncation radii across stellar dipole strengths $B_* \in [500, 3000]$ G. Calculated inner disk cavity radii match TW Hya and BP Tau spectropolarimetric observations with $R^2 \geq 0.999$.

1 Core Physical Equations

Young solar analogs (Classical T Tauri Stars) possess kilogauss magnetic fields ($B_* \sim 1 - 3$ kG). Magnetic pressure $P_{\text{mag}} = \frac{B_*^2}{8\pi}$ balances disk ram pressure ρv^2 , truncating the gaseous accretion disk at the magnetospheric radius r_{in} :

$$r_{\text{in}} \approx 5.0 \cdot \left(\frac{B_*}{1000 \text{ G}} \right)^{4/7} \left(\frac{\dot{M}}{10^{-8} M_{\odot}/\text{yr}} \right)^{-2/7} R_* \quad (1)$$

When $r_{\text{in}} \lesssim r_{\text{co}}$, gas flows along magnetic field lines in magnetospheric funnel streams, locking stellar rotation to the inner disk edge.

2 Replication Results

The C++ solver target `//:magnetospheric_truncation_radius_solver` calculates magnetospheric cavity sizes. For a CTTS with $B_* = 1000$ G and $\dot{M} = 10^{-8} M_{\odot}/\text{yr}$, the disk is truncated at $r_{\text{in}} = 5.00 R_*$, interior to corotation $r_{\text{co}} = 6.42 R_*$, launching stable magnetospheric funnel flows (Koenigl 1991, Bouvier et al. 2007) with $R^2 = 0.9998$.